

Aman Yadav

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SUMMARY

Mathematics & Computing undergraduate architecting scalable, high-concurrency backend services, distributed asynchronous worker pipelines, and low-latency machine learning inference microservices. Hands-on experience designing robust relational schemas, message queues, and production APIs utilizing FastAPI, PostgreSQL, Celery/Redis, and Docker.

EDUCATION

Central University of Karnataka

Bachelor of Technology (B.Tech) in Mathematics & Computing

Kalaburagi, Karnataka, India

Expected 2027

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Database Management Systems (DBMS), Distributed Computing, Object-Oriented Programming, Linear Algebra, Probability & Statistics, Numerical Optimization.

TECHNICAL EXPERIENCE

Maruti Suzuki India Limited

Engineering Intern (Applied ML & Telematics) – EN-DAX Dept

20 May 2026 – 15 July 2026

Gurugram, India

- **Telemetry Ingestion & Standardization:** Architected an industrial **6-stage telemetry pipeline** normalizing raw high-frequency **CAN-bus sensor feeds** into a **40-feature automotive taxonomy** with automated trip segmentation.
- **Kinematic Derivations & Feature Engineering:** Formulated trajectory modeling using vectorized **Haversine calculations**, continuous speed differentials, and threshold-based triggers to detect **aggressive driving maneuvers** (hard braking, rapid acceleration).
- **Statistical Anomaly Cleansing:** Implemented dual-stage **IQR and Z-score outlier suppression** pipelines, filtering sensor noise and telemetry drift while persisting structured **JSON validation audit logs**.
- **Internal Analytics Console:** Deployed an interactive **Streamlit engineering console** for vehicle dynamics engineers to visualize sensor time-series profiles, trip heatmaps, and kinematic distributions in real time.

FEATURED PROJECTS

PingGuard — Distributed HTTP Uptime Monitoring Engine



- **Decoupled Worker Architecture:** Engineered an asynchronous monitoring service separating the **FastAPI control plane** from outbound probing workloads via **Celery worker fleets** backed by **Redis**.
- **Concurrency & Task Dispatching:** Implemented database-driven worker dispatching using **PostgreSQL row-level locking** (`SELECT ... FOR UPDATE SKIP LOCKED`) to **eliminate race conditions** and redundant task execution.
- **Network Security Hardening:** Implemented pre-flight **asynchronous DNS validation** against private CIDR blocklists and direct IP pinning to prevent **TOCTOU DNS rebinding** and **SSRF exploits**.

EduPulse — Academic Forecasting & Early-Warning Platform



- **Enterprise Backend & RBAC:** Architected a modular RESTful backend with **Role-Based Access Control (RBAC)**, **JWT authentication**, and transactional **PostgreSQL** persistence.
- **Isolated Caching & Queue Broker:** Designed a **dual-Redis infrastructure** separating memory caching from Celery task queues, utilizing **generational cache invalidation** to achieve **<40ms response times**.
- **Predictive ML Inference:** Integrated baseline prior and longitudinal institutional models with **GroupedKFold cross-validation** and demographic quarantine to ensure **zero PII leakage** during automated batch evaluation.

EV-Lifespan — EV Battery Degradation & RUL Platform



- **Low-Latency Inference Service:** Developed a high-throughput **FastAPI inference microservice** serving PyTorch sequence models with **<15ms latency** under continuous multi-channel sensor feeds.
- **Spatial-Temporal Neural Architecture:** Engineered a 2-tier prognostic engine combining **1D-CNN spatial feature extractors** (voltage/current/temperature profiles) with **stacked LSTM sequence models**, achieving **19.40 RMSE cycles**.
- **Containerization & Production Packaging:** Built **multi-stage Docker container** environments with automated health checks, unified API contracts, and real-time degradation curve streaming.

TECHNICAL SKILLS

Languages: Python, SQL (PostgreSQL), C++, TypeScript, Bash

Backend & Distributed Systems: FastAPI, Django, Celery, Redis, RESTful APIs, AsyncIO, WebSockets, JWT Auth, Microservices

Databases & ORM: PostgreSQL, Redis (Caching & Queues), SQLAlchemy, Alembic, Relational Schema Design

DevOps & Infrastructure: Docker, Docker Compose, Git, CI/CD Pipelines, Postman, Render, Streamlit, Vercel

Applied ML & Quantitative: PyTorch, Scikit-Learn, Pandas, NumPy, Linear Algebra, Time-Series Modeling